Definite Integrals: Examples Worksheet Solutions

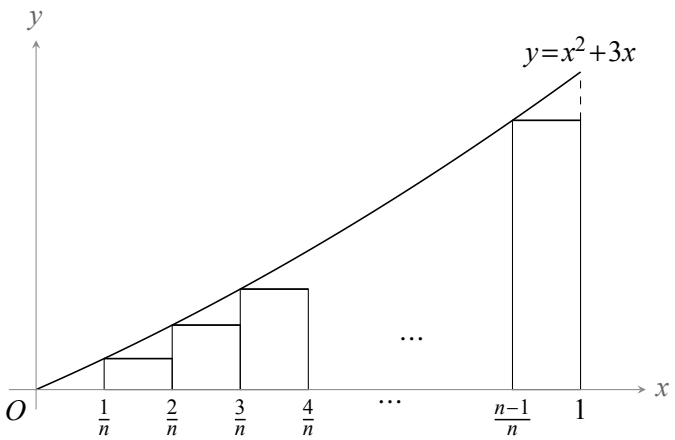
Compiled by Maximus Pung Jun

JC and A-level examination questions (H2 Mathematics 9758): three on the limit of a sum for Section 10.1, then one for each later section (Promotional Examination where available, otherwise Preliminary Examination). Questions are reproduced as set, with the original marks.

Q	Section	Source
1	10.1 The definite integral as a limit of a sum	DHS/2019/P1/Q10†
2	10.1 The definite integral as a limit of a sum	ACJC/2019/P1/Q6†
3	10.1 The definite integral as a limit of a sum	9758/2015/P1/Q3†
4	10.2 Area between a curve and the x-axis	NJC 2024 Promo Q3
5	10.3 Area between a curve and the y-axis	SAJC 2025 Prelim P1 Q7
6	10.4 Area between two curves	YIJC 2025 Prelim P1 Q12
7	10.5 Volume of revolution	NJC 2024 Promo Q2
8	10.6 Parametric curves: area	VJC 2025 Promo Q11(c), adapted

†From the Section 1 exercises in the teacher's own Definite Integrals notes.

1 10.1 [DHS/2019/P1/Q10]



The diagram above shows part of the graph of $y=x^2+3x$, with rectangles approximating the area under the curve from $x=0$ to $x=1$. The area under the curve may be approximated by the total area, A, of $(n-1)$ rectangles each of width $\frac{1}{n}$. Given that

$$\sum_{r=1}^{n} r^2=\frac{1}{6}n(n+1)(2n+1),$$

show that $A=\frac{(n-1)(11n-1)}{6n^2}$.

Explain briefly how the value of $\int_0^1 x^2+3x\ dx$ can be deduced from this expression, and hence find this value exactly without integration. **[6]**

Solution

With $f(x)=x^2+3x$:

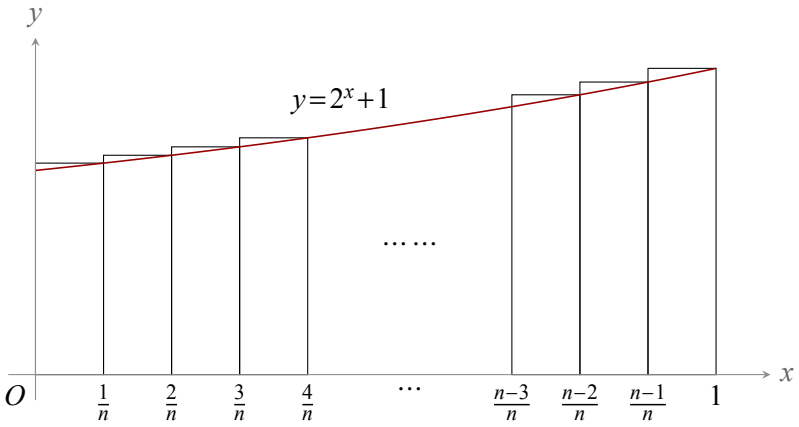
$$\begin{aligned}
A &= \frac{1}{n}\left[\left(\frac{1}{n}\right)^2+3\left(\frac{1}{n}\right)\right]+\frac{1}{n}\left[\left(\frac{2}{n}\right)^2+3\left(\frac{2}{n}\right)\right]+\frac{1}{n}\left[\left(\frac{3}{n}\right)^2+3\left(\frac{3}{n}\right)\right] \\
&\quad +\cdots+\frac{1}{n}\left[\left(\frac{n-1}{n}\right)^2+3\left(\frac{n-1}{n}\right)\right] \\
&= \sum_{r=1}^{n-1}\frac{1}{n}\left[\left(\frac{r}{n}\right)^2+3\left(\frac{r}{n}\right)\right] \\
&= \sum_{r=1}^{n-1}\left(\frac{1}{n^3}r^2+\frac{3}{n^2}r\right) \\
&= \frac{1}{n^3}\sum_{r=1}^{n-1}r^2+\frac{3}{n^2}\sum_{r=1}^{n-1}r \\
&= \frac{1}{n^3}\left[\frac{(n-1)(n)(2n-1)}{6}\right]+\frac{3}{n^2}\left[\frac{n(n-1)}{2}\right] \\
&= \frac{(n-1)(2n-1)}{6n^2}+\frac{3(n-1)}{2n} \\
&= \frac{(n-1)(2n-1)+9n(n-1)}{6n^2} \\
&= \frac{(n-1)(2n-1+9n)}{6n^2} \\
&= \frac{(n-1)(11n-1)}{6n^2} \text{ (shown)}
\end{aligned}$$

As $n \to \infty$, the total area of the rectangles approaches the actual area under the curve, which is $\int_0^1 (x^2+3x)\,dx$.

$$\begin{aligned}
\int_0^1 (x^2+3x)\,dx &= \lim_{n\to\infty}\frac{(n-1)(11n-1)}{6n^2} \\
&= \lim_{n\to\infty}\frac{11n^2-12n+1}{6n^2} \\
&= \lim_{n\to\infty}\frac{11-\frac{12}{n}+\frac{1}{n^2}}{6} \\
&= \frac{11}{6}
\end{aligned}$$

2 **10.1** [ACJC/2019/P1/Q6]

The diagram below shows the graph of $y=2^x+1$ for $0\le x\le 1$. Rectangles, each of width $\frac{1}{n}$, are also drawn on the graph as shown.



Show that the total area of all n rectangles, S_n, is given by

$$S_n=\frac{2^{1/n}}{n\left(2^{1/n}-1\right)}+1.$$

[3]

Find the exact value of $\lim_{n\to\infty} S_n$. **[2]**

Solution

Show that:

$$\begin{aligned}
S_n&=\frac{1}{n}\left(2^{\frac{1}{n}}+1\right)+\frac{1}{n}\left(2^{\frac{2}{n}}+1\right)+\frac{1}{n}\left(2^{\frac{3}{n}}+1\right)\\
&\quad+\cdots+\frac{1}{n}\left(2^{\frac{n}{n}}+1\right)\\
&=\sum_{r=1}^{n}\frac{1}{n}\left(2^{\frac{r}{n}}+1\right)\\
&=\frac{1}{n}\sum_{r=1}^{n}2^{\frac{r}{n}}+\frac{1}{n}\sum_{r=1}^{n}1\\
&=\frac{1}{n}\sum_{r=1}^{n}\left(2^{\frac{1}{n}}\right)^r+\frac{1}{n}(n)\\
&=\frac{1}{n}\left[\frac{2^{\frac{1}{n}}\left[\left(2^{\frac{1}{n}}\right)^n-1\right]}{2^{\frac{1}{n}}-1}\right]+1\\
&=\frac{1}{n}\left[\frac{2^{\frac{1}{n}}(2-1)}{2^{\frac{1}{n}}-1}\right]+1\\
&=\frac{2^{\frac{1}{n}}}{n\left(2^{\frac{1}{n}}-1\right)}+1 \text{ (shown)}
\end{aligned}$$

Sum of a GP: $\sum_{r=1}^{n}a^r=\frac{a\left(a^n-1\right)}{a-1}$

Alternatively: $2^{\frac{1}{n}}$, $2^{\frac{2}{n}}$, $2^{\frac{3}{n}}$, ... is a GP; each term is $2^{\frac{1}{n}}$ times the one before.

$$\begin{aligned}
S_n&=\tfrac{1}{n}\left(2^{\frac{1}{n}}+2^{\frac{2}{n}}+2^{\frac{3}{n}}+\cdots+2^{\frac{n}{n}}\right)\\
&\quad+\tfrac{1}{n}(1+1+1+\cdots+1)\\
&=\tfrac{1}{n}\left(\frac{2^{\frac{1}{n}}\left[\left(2^{\frac{1}{n}}\right)^n-1\right]}{2^{\frac{1}{n}}-1}\right)+\tfrac{1}{n}(n)\\
&=\vdots \quad \text{(as on the left)}
\end{aligned}$$

Limit:

$$\begin{aligned}
\lim_{n\to\infty}S_n&=\int_0^1(2^x+1)\,dx\\
&=\left[\frac{2^x}{\ln 2}+x\right]_0^1\\
&=\frac{2}{\ln 2}+1-\frac{1}{\ln 2}\\
&=\frac{1}{\ln 2}+1
\end{aligned}$$

3 **10.1** [9758/2015/P1/Q3]

(i) Given that f is a continuous function, explain, with the aid of a sketch, why the value of

$$\lim_{n\to\infty}\frac{1}{n}\left\{\mathrm{f}\left(\frac{1}{n}\right)+\mathrm{f}\left(\frac{2}{n}\right)+\cdots+\mathrm{f}\left(\frac{n}{n}\right)\right\}$$

is $\int_0^1 \mathrm{f}(x)\,dx$. **[2]**

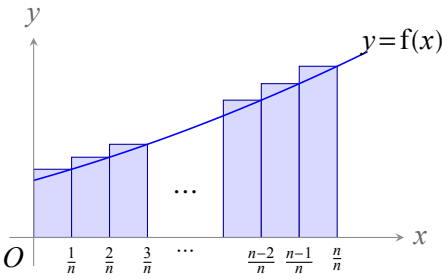
(ii) Hence evaluate $\lim_{n\to\infty}\frac{1}{n}\left(\frac{\sqrt[3]{1}+\sqrt[3]{2}+\cdots+\sqrt[3]{n}}{\sqrt[3]{n}}\right)$. **[3]**

Solution

(i)

Total area of rectangles

$$=\frac{1}{n}\mathrm{f}\left(\frac{1}{n}\right)+\frac{1}{n}\mathrm{f}\left(\frac{2}{n}\right)+\frac{1}{n}\mathrm{f}\left(\frac{3}{n}\right)+\cdots+\frac{1}{n}\mathrm{f}\left(\frac{n}{n}\right)$$
$$=\frac{1}{n}\left[\mathrm{f}\left(\frac{1}{n}\right)+\mathrm{f}\left(\frac{2}{n}\right)+\mathrm{f}\left(\frac{3}{n}\right)+\cdots+\mathrm{f}\left(\frac{n}{n}\right)\right]$$



As $n\to\infty$, the total area of the rectangles approaches the actual area under the graph, which is $\int_0^1 \mathrm{f}(x)\,dx$.

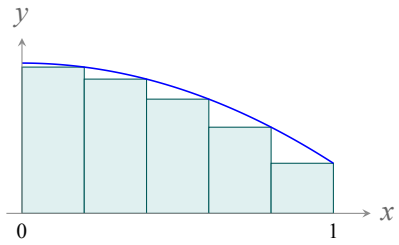
$$\therefore \lim_{n\to\infty}\frac{1}{n}\left[\mathrm{f}\left(\frac{1}{n}\right)+\mathrm{f}\left(\frac{2}{n}\right)+\mathrm{f}\left(\frac{3}{n}\right)+\cdots+\mathrm{f}\left(\frac{n}{n}\right)\right]=\int_0^1 \mathrm{f}(x)\,dx \text{ (shown)}$$

(ii)

$$\lim_{n\to\infty}\frac{1}{n}\left(\sqrt[3]{\frac{1}{n}}+\sqrt[3]{\frac{2}{n}}+\sqrt[3]{\frac{3}{n}}+\cdots+\sqrt[3]{\frac{n}{n}}\right)=\int_0^1 \sqrt[3]{x}\,dx$$
$$=\int_0^1 x^{\frac{1}{3}}\,dx$$
$$=\left[\frac{x^{\frac{4}{3}}}{\frac{4}{3}}\right]_0^1$$
$$=\frac{3}{4}$$

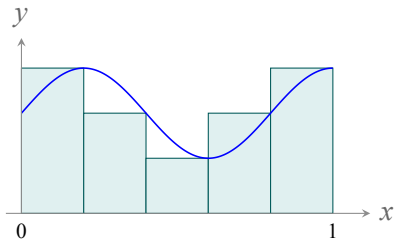
Question 3: right-hand or left-hand rectangles? (5 rectangles on $[0,1]$)

Decreasing, right-hand ✓



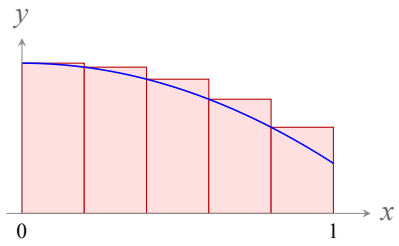
rectangles = 1.06, area = 1.167: underestimate

Up and down, right-hand ✓



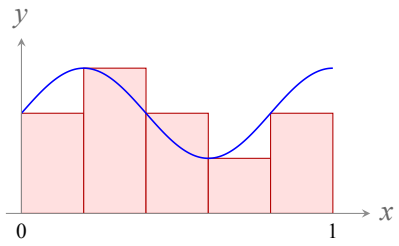
rectangles = 1.09, area = 1.057: overestimate

Decreasing, left-hand ✗



rectangles = 1.26, area = 1.167: overestimate

Up and down, left-hand ✗



rectangles = 1.00, area = 1.057: underestimate

What matters is that the rectangles use the **right-hand sum** $\frac{1}{n}\left[\mathrm{f}\left(\frac{1}{n}\right)+\mathrm{f}\left(\frac{2}{n}\right)+\cdots+\mathrm{f}\left(\frac{n}{n}\right)\right]$, as the question specifies, not whether it is an under- or overestimate. It is not correct to say this question only accepts overestimates: a right-hand sum alone does not determine under or over (top row), and a left-hand sum $\frac{1}{n}\left[\mathrm{f}\left(\frac{0}{n}\right)+\cdots+\mathrm{f}\left(\frac{n-1}{n}\right)\right]$ is not what the question describes, even when it overestimates (bottom row).

4 **10.2** [NJC 2024 Promo Q3]

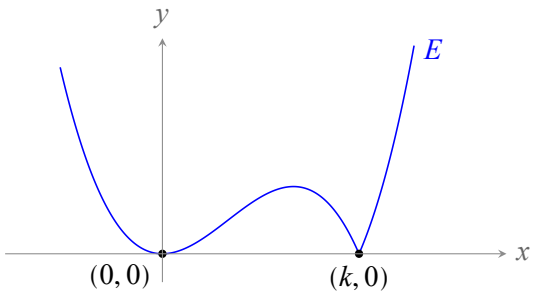
(i) Given that k is a positive constant, sketch the curve E with equation $y=|x^2(x-k)|$, labelling the coordinates of the points where the curve cuts the axes. **[2]**

(ii) Hence, find, in terms of k, the area of the region(s) bounded by E and the x-axis for $0 \le x \le 2k$. **[3]**

Solution

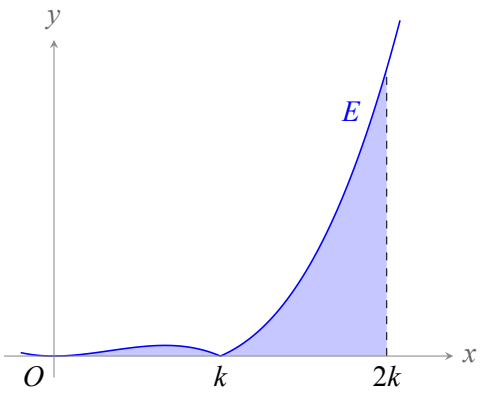
Working follows the school's solution on grail.moe: https://grail.moe/library/16719/njc-9758-2024-promo-solutions

(i)



(ii)

$$\begin{aligned}
&\int_0^{2k} |x^2(x-k)|\,dx \\
&= \int_0^{2k} |x^3 - kx^2|\,dx \\
&= -\int_0^{k} x^3 - kx^2\,dx + \int_k^{2k} x^3 - kx^2\,dx \\
&= -\left[\frac{x^4}{4} - k\frac{x^3}{3}\right]_0^{k} + \left[\frac{x^4}{4} - k\frac{x^3}{3}\right]_k^{2k} \\
&= \frac{3}{2}k^4
\end{aligned}$$



5 **10.3** [SAJC 2025 Prelim P1 Q7]

(a) (i) On a single diagram, sketch the graphs of $y=\dfrac{3x}{x-1}$ and $y=3x$, stating the equation(s) of any asymptote(s) and the coordinates of points of intersection. **[4]**

(ii) Hence find the area bounded by the curve $y=\dfrac{3x}{x-1}$ and the lines $y=3x$ and $y=8$. **[2]**

(b) A curve has the equation $y=\sqrt{x^2+1}$ where x is non-negative.

(i) With the aid of a sketch of the above curve, explain why the value of

$$S=\lim_{n\to\infty}\frac{1}{n}\left[1+\sqrt{\frac{1}{n^2}+1}+\sqrt{\frac{4}{n^2}+1}+\ldots+\sqrt{\frac{(n-1)^2}{n^2}+1}\right] \text{ is } \int_0^1\sqrt{x^2+1}\,dx.$$

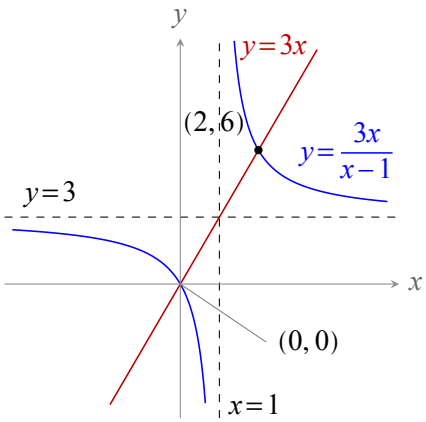
[4]

(ii) Hence find the value of S. **[1]**

Solution

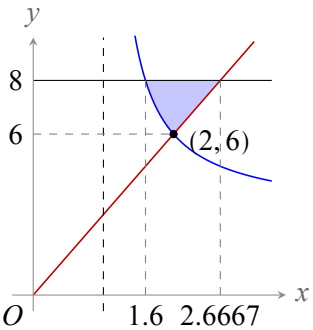
Working follows the school's solution on grail.moe: https://grail.moe/library/19502/sajc-9758-2025-prelim-p1-solutions

(a)(i)



(a)(ii)

Area of region bounded by both curves and $y=8=\int_6^8\left(\dfrac{y}{3}-\dfrac{y}{y-3}\right)dy$



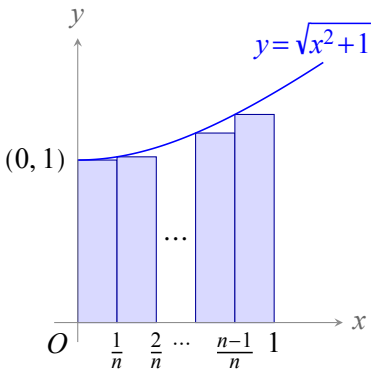
Using GC,

$$\text{Area}=1.13 \text{ units}^2 \text{ (3 s.f.)}$$

Alternative method:

$$\begin{aligned}\text{Area}&=(2.6667-1.6)(8)-\int_{1.6}^{2}\frac{3x}{x-1}\,dx-\int_{2}^{2.6667}3x\,dx\\&=8.5336-2.7325-4.6669\\&=1.13 \text{ units}^2 \text{ (3 s.f.)}\end{aligned}$$

(b)(i) If we split the area of the region enclosed by the curve $y=\mathrm{f}(x)=\sqrt{x^2+1}$, the x-axis and the lines $x=0$ and $x=1$ into n rectangles, each of width $\frac{1}{n}$ as shown, the sum of the area of the n rectangles is given by



$$\begin{aligned}A&=\frac{1}{n}\mathrm{f}(0)+\frac{1}{n}\mathrm{f}\left(\frac{1}{n}\right)+\frac{1}{n}\mathrm{f}\left(\frac{2}{n}\right)+\cdots+\frac{1}{n}\mathrm{f}\left(\frac{n-1}{n}\right)\\&=\frac{1}{n}\left[\mathrm{f}(0)+\mathrm{f}\left(\frac{1}{n}\right)+\mathrm{f}\left(\frac{2}{n}\right)+\cdots+\mathrm{f}\left(\frac{n-1}{n}\right)\right]\\&=\frac{1}{n}\left[1+\sqrt{\frac{1}{n^2}+1}+\sqrt{\frac{4}{n^2}+1}+\cdots+\sqrt{\frac{(n-1)^2}{n^2}+1}\right]\end{aligned}$$

As the number of rectangles increases, i.e. $n\to\infty$, the width of each rectangle decreases, then the sum of the area of the n rectangles approaches the area of the region enclosed by the curve $y=\sqrt{x^2+1}$, the x-axis and the lines $x=0$ and $x=1$, which is given by $\int_0^1\sqrt{x^2+1}\,dx$. Hence

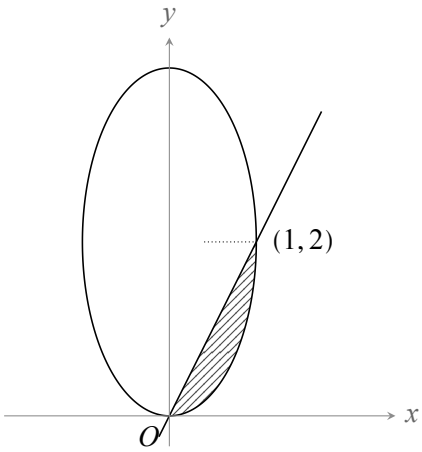
$$\begin{aligned}S&=\lim_{n\to\infty}A\\&=\int_0^1\mathrm{f}(x)\,dx\\&=\int_0^1\sqrt{1+x^2}\,dx\end{aligned}$$

(b)(ii) Actual area under the graph $y=\sqrt{x^2+1}$ from $x=0$ to $x=1$. Using GC,

$$\begin{aligned}\int_0^1\sqrt{x^2+1}\,dx&=1.1478\\&=1.15\text{ (3 s.f.)}\end{aligned}$$

6 **10.4** [YIJC 2025 Prelim P1 Q12]

(a) Use the substitution $x=\sin\theta$ to show that $\int_0^1 \sqrt{1-x^2}\,dx=\frac{\pi}{4}$. **[4]**



The diagram shows the region R bounded by the ellipse with equation $4x^2+(y-2)^2=4$ and the line $y=2x$. The ellipse and the line intersect at the origin and the point $(1,2)$.

(b) Find the area of R, giving your answer in terms of π. **[4]**

(c) Find the volume of solid generated when R is rotated 2π radians about the x-axis. **[2]**

Solution

Working follows the school's solution on grail.moe: https://grail.moe/library/18935/2025-yijc-prelim-h2-maths-papers-1-2-qp-soln

(a)

$$x=\sin\theta$$
$$\frac{dx}{d\theta}=\cos\theta$$

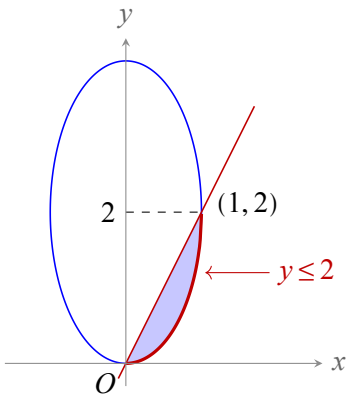
When $x=0,\ \theta=0;\ x=1,\ \theta=\frac{\pi}{2}$

$$\begin{aligned}\int_0^1 \sqrt{1-x^2}\,dx&=\int_0^{\frac{\pi}{2}} \sqrt{1-\sin^2\theta}\,\cos\theta\,d\theta\\&=\int_0^{\frac{\pi}{2}} \cos^2\theta\,d\theta\\&=\int_0^{\frac{\pi}{2}} \frac{1}{2}+\frac{1}{2}\cos 2\theta\,d\theta\\&=\left[\frac{1}{2}\theta+\frac{1}{4}\sin 2\theta\right]_0^{\frac{\pi}{2}}\\&=\left[\frac{1}{2}\left(\frac{\pi}{2}\right)+\frac{1}{4}\sin\pi\right]-\left[\frac{1}{2}(0)+\frac{1}{4}\sin 0\right]\\&=\frac{\pi}{4}\end{aligned}$$

! Remember to change the limits when doing integration by substitution.
To get rid of the square root, use $1-\sin^2\theta=\cos^2\theta$ (trigo identity).
Use MF27 (double angle formula) $\cos 2\theta=2\cos^2\theta-1$ to integrate $\cos^2\theta$.

(b)

$$4x^2+(y-2)^2=4$$
$$(y-2)^2=4-4x^2$$
$$y-2=\pm\sqrt{4-4x^2}$$
$$y=2\pm2\sqrt{1-x^2}$$



Since $y\le 2$, $y=2-2\sqrt{1-x^2}$

$$\text{Area of } R=\frac{1}{2}(2)(1)-\int_0^1 y\,dx$$
$$=1-\int_0^1 2-2\sqrt{1-x^2}\,dx$$
$$=1-\int_0^1 2\,dx+2\int_0^1 \sqrt{1-x^2}\,dx$$
$$=1-\left[2x\right]_0^1+2\left(\frac{\pi}{4}\right)$$
$$=\frac{\pi}{2}-1$$

Alternatively

$$\text{Area of } R=\int_0^1 2x-\left(2-2\sqrt{1-x^2}\right)dx$$

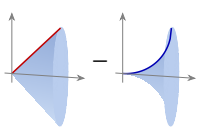
("upper curve" – "lower curve")

! Observe from the graph that the y values on the curve that we are using to find the area under the curve are less than or equal to 2. Thus, we choose $y=2-2\sqrt{1-x^2}$.

(c)

$$\text{Volume}=\frac{1}{3}\pi(2)^2(1)-\pi\int_0^1\left(2-2\sqrt{1-x^2}\right)^2dx$$
$$=2.98 \text{ (3 s.f.)}$$

(c)



Alternatively

$$\text{Volume}=\pi\int_0^1(2x)^2\,dx-\pi\int_0^1\left(2-2\sqrt{1-x^2}\right)^2dx$$

! Required volume = Volume of cone − Volume generated by the region under the curve.
Volume of cone can be found by using $\frac{1}{3}\pi r^2h$ or $\pi\int_0^1(2x)^2\,dx$.

7 10.5

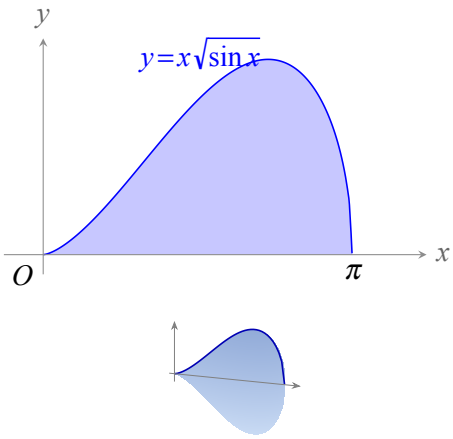
[NJC 2024 Promo Q2]

The region bounded by $y=x\sqrt{\sin x}$, the x-axis, and the lines $x=0$ and $x=\pi$ is rotated through 2π radians about the x-axis. Find the exact volume of the solid generated. **[5]**

Solution

Working follows the school's solution on grail.moe: https://grail.moe/library/16719/njc-9758-2024-promo-solutions

$$\begin{aligned}\int y^2\,dx &= \int x^2\sin x\,dx\\ &= -x^2\cos x-\int 2x(-\cos x)\,dx\\ &= -x^2\cos x+2\int x(\cos x)\,dx\\ &= -x^2\cos x+2\left[x\sin x-\int \sin x\,dx\right]\\ &= -x^2\cos x+2\left[x\sin x+\cos x\right]\\ &= -x^2\cos x+2x\sin x+2\cos x\\ &= \left(2-x^2\right)\cos x+2x\sin x+c\end{aligned}$$



$$\begin{aligned}\text{Volume required} &= \pi\int_0^{\pi} y^2\,dx\\ &= \pi\left[\left(2-x^2\right)\cos x+2x\sin x\right]_0^{\pi}\\ &= \pi\left[\left(2-\pi^2\right)(-1)-(2)1\right]\\ &= \pi^3-4\pi\end{aligned}$$

8 **10.6** [VJC 2025 Promo Q11(c), adapted]

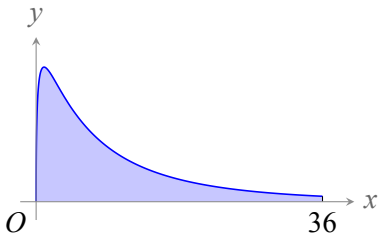
The parametric equations $x=t^2$, $y=20te^{-t}$, for $0\le t\le 6$, model a section of a roller coaster track; x is the horizontal distance and y the vertical distance from the ground, in metres. A wind-blocking panel is attached beneath the track. Its cross-sectional area is modelled by the area under the curve from $t=0$ to $t=6$. The area of the region under a parametric curve from $t=\alpha$ to $t=\beta$ is given by the formula $\int_{\alpha}^{\beta} y\left(\frac{dx}{dt}\right)dt$.

(i) Show that the cross-sectional area of the panel is $\int_a^b kt^2e^{-t}\,dt$, where a, b and k are integers to be determined. **[2]**

(ii) Find the exact area of the panel. **[4]**

Solution

(i)



$$\frac{dx}{dt}=2t$$

$$\text{Area}=\int_0^6 y\left(\frac{dx}{dt}\right)dt$$

$$=\int_0^6 (20te^{-t})\,(2t)\,dt$$

$$=\int_0^6 40t^2e^{-t}\,dt$$

$$\therefore k=40,\quad a=0,\quad b=6 \text{ (shown)}$$

(ii)

$$\int t^2e^{-t}\,dt=-\int t^2(-1)e^{-t}\,dt$$

$$=-t^2e^{-t}-\int(2t)\,(-e^{-t})\,dt$$

$$=-t^2e^{-t}+\int 2te^{-t}\,dt$$

$$=-t^2e^{-t}-2te^{-t}-\int(2)\,(-e^{-t})\,dt$$

$$=-t^2e^{-t}-2te^{-t}+\int 2e^{-t}\,dt$$

$$=-t^2e^{-t}-2te^{-t}-2\int(-1)e^{-t}\,dt$$

$$=-t^2e^{-t}-2te^{-t}-2e^{-t}+C$$

$$=-e^{-t}\left(t^2+2t+2\right)+C$$

$$\text{Area}=40\int_0^6 t^2e^{-t}\,dt$$

$$=40\left[-e^{-t}\left(t^2+2t+2\right)\right]_0^6$$

$$=40\left[-e^{-6}\,(36+12+2)-\left(-e^{0}(0+0+2)\right)\right]$$

$$=40\left(-50e^{-6}+2\right)$$

$$=80-2000e^{-6}\text{ m}^2$$